



SUJET STATAPP 303 - 2026-2027



Sujet proposé par : Guillot Gilles - g.guillot@woah.org
Organisme : World Organisation for Animal Health (WOAH)
Catégorie : Traitement du signal / NLP
Sujet :
Embeddings for Epidemic Intelligence

Description du sujet :

Context

WOAH's Epidemic Intelligence (EI) team within the Data Integration and Analytics Department, monitors open-source media (news wires, health-authority bulletins, agricultural press, social media) to detect early, unconfirmed ("weak") signals of animal disease events worldwide, ahead of and alongside official notification through the WOAH World Animal Health Information System (WAHIS). This scanning uses automated information-acquisition, information-retrieval and text-analysis approaches ? in the same family as PADI-web (CIRAD/ANSES/WOAH), HealthMap, GPHIN, BioCaster or WHO's EIOS ? that flag candidate articles chiefly through **lexical matching**: curated keyword lists, Boolean queries and taxonomies of disease/host/symptom terms, in some cases combined with a supervised classifier trained on bag-of-words or n-gram features.

Rationale

Lexical matching has two well-documented weaknesses for epidemic intelligence. First, **recall**: an article that reports a disease event but contains none of the listed trigger terms (a synonym, a local name, a vague clinical description, a paraphrase in a language the keyword list wasn't built for) is missed outright ? this is the classic vocabulary-mismatch problem in information retrieval. Second, **precision**: keyword lists trigger on off-topic uses of the same terms (e.g. a "swine fever" reference in a historical or fictional context, or a keyword that is also a place or company name), which pushes human curation time toward triage rather than analysis. Both failure modes scale badly as the set of monitored languages and diseases grows.

Dense text **embeddings** ? vector representations that place semantically similar text close together regardless of exact wording or, for multilingual models, of language ? are a natural candidate to complement or replace keyword triggers with semantic similarity search: retrieve or classify articles by how close their embedding is to a small set of reference descriptions of a disease event, rather than by whether they contain a specific string. The approach is already established in general-purpose semantic search and open-domain question answering; it has seen comparatively little systematic evaluation as a front end for animal-health event-based surveillance, where the operational bar is not just top-k retrieval quality but a documented, defensible sensitivity and specificity against the kind of confirmed-event data (WAHIS) and expert-annotated corpora the EI team already curates.

This student project develops a proof of concept for this comparison: a semantic-matching pipeline that could serve as an alternative or complement to lexical-based retrieval, evaluated against the same benchmark and under the same conditions as the existing approaches.

Données du sujet :

Reference event / calibration data

- **WOAH WAHIS** ? confirmed disease events (location, date, disease, host) usable as a (partial, delayed) gold standard against which candidate signals can be scored.
- **PADI-web** (CIRAD/ANSES, operated in support of WOA's EI activities) ? published methodology and, where accessible, gold-standard annotated article sets used in its own validation studies; a natural lexical baseline to benchmark against.
- **HealthMap** and **WHO EIOS** ? published methodology and, where available, sample outputs, as further points of comparison for an event-based-surveillance baseline.
- **Domain corpora for embedding fine-tuning/evaluation**: PubMed/PMC abstracts (veterinary and zoonotic disease subset), FAO/WOA technical bulletins, WOA disease factsheets.
- Expert-annotated corpora from the epidemic-intelligence NLP literature (e.g. the PADI-web and BioCaster gold-standard sets described in the literature below), where reusable under their stated licence.

Text corpora for pipeline development

- Multilingual news aggregators (e.g. EIOS, Padi-web, GDELT, NewsAPI-type feeds) to build the raw scraped-article stream the pipeline will operate on, mirroring what the EI team's own tools ingest.
- Domain-specific corpora for embedding evaluation and, where appropriate, domain adaptation, including PubMed/PMC abstracts relevant to veterinary and zoonotic diseases, FAO and WOA technical publications and bulletins, and WOA disease factsheets. These resources can be used to assess whether semantic representations capture relevant disease-, host-, and event-related concepts.
- Independent evaluation corpus: where feasible, construct a manually annotated sample of news articles to establish a gold standard for disease-event relevance. This enables precision, recall and F1 to be compared between lexical and semantic approaches under the same conditions.

Embedding models

- General-purpose multilingual sentence embeddings: LaBSE, multilingual-E5 / BGE-M3, LASER.
- English and biomedical-domain embeddings: Sentence-BERT variants, BioBERT, PubMedBERT.
- Commercial/API embeddings (OpenAI, Cohere multilingual) as a proprietary-model comparison point, budget permitting.
- Classic distributional baselines (word2vec, GloVe, fastText) as a lower-complexity reference point against transformer-based sentence embeddings.

Modèles envisagés du sujet :

Tasks foreseen in this internship include:

- **Review of embedding models**: survey the candidate model families above, focusing on (i) multilingual coverage matching the languages the EI team actually monitors, (ii) domain/vocabulary fit for animal-health and veterinary terminology, (iii) inference cost and latency at web-scraping scale, and (iv) licensing/deployability constraints (self-hosted vs. API).
- **Implementation of a data pipeline based on various embeddings**: develop a semantic-matching component using at least two or three embedding models, representing incoming articles and a set of reference disease-event descriptions and flagging candidate articles based on semantic similarity (e.g. nearest-neighbour or threshold-based retrieval). The pipeline should be designed to operate both on the existing EIOS-derived information stream and, where feasible, on an independent stream of articles from the same or comparable open-source sources. This will allow the project to distinguish between (i) the added value of semantic matching for prioritising information already retrieved by EIOS and (ii) its ability to identify relevant articles that may be missed by the existing keyword-based retrieval process.
- **Assessment of sensitivity and specificity of the pipelines, accounting for error in the calibrating data**: benchmark each pipeline's detection performance against WAHIS-confirmed events and/or expert-annotated article sets; because

the reference ("gold standard") data is itself imperfect ? WAHIS under-reports and lags, expert annotation carries inter-annotator disagreement ? use latent-class / no-gold-standard estimation methods (in the tradition of Hui?Walter and Rogan?Gladen models, standard in veterinary diagnostic-test evaluation) to correct sensitivity/specificity estimates for calibration-data error rather than treating the reference data as ground truth.

- **Comparison with the existing lexical baseline**: apply the same evaluation protocol to the current lexical/keyword-based approach used for EIOS-based epidemic intelligence, or to a reproducible approximation based on established methodologies such as PADI-web. Quantify where semantic matching improves or reduces recall and precision, and identify the languages, diseases and types of signals for which semantic approaches provide the greatest added value.

- **Communication, interaction and coordination with the EI team**: maintain regular interaction with the Epidemic Intelligence team within the Data Integration and Analytics Department throughout the project, including discussion of pipeline design, data requirements, evaluation results and interpretation of findings, and communicate the final results and recommendations to the team.

Bibliographie du sujet :

Event-based epidemic intelligence: tools and systems

- **[E]** Brownstein, Freifeld & Madoff (2008), "Digital Disease Detection ? Harnessing the Web for Public Health Surveillance," New England Journal of Medicine. ? Founding description of HealthMap-style automated, keyword-driven web surveillance and its role alongside traditional reporting.

- **[E]** Mykhalovskiy & Weir (2006), "The Global Public Health Intelligence Network and Early Warning Outbreak Detection," Canadian Journal of Public Health. ? Describes GPHIN's keyword/rule-based multilingual media-monitoring approach and its role in the SARS response; an early articulation of the lexical-matching paradigm this project seeks to complement.

- **[E]** Collier et al. (2008), "BioCaster: Detecting Public Health Rumors with a Web-Based Text Mining System," Bioinformatics. ? Ontology- and rule-based text mining for outbreak rumour detection; illustrates the taxonomy/keyword architecture common to this generation of tools.

- **[G]** Madoff (2004), "ProMED-mail: An Early Warning System for Emerging Diseases," Clinical Infectious Diseases. ? Describes the human-curated free-text reporting model that remains a reference corpus for outbreak narrative text.

- **[E]** Arsevska et al. (2018), "Web Monitoring of Emerging Animal Infectious Diseases Integrated in the French Animal Health Epidemic Intelligence System," PLOS ONE. ? Describes PADI-web's keyword-query and supervised-classification pipeline for animal-disease event detection from online news, including its validation against expert-confirmed events; the direct lexical-matching baseline for this project.

- **[E]** Valentin et al. (2020), "PADI-web 3.0: Automatic Monitoring of Health Threats through Online News for Enhanced Epidemic Intelligence," Ecological Informatics. ? Later iteration adding automatic event extraction; useful methodological detail on how keyword-triggered candidate articles are subsequently filtered and structured.

- **[G]** WHO, "Epidemic Intelligence from Open Sources (EIOS)" ? institutional description of the multi-source, multilingual media-monitoring platform used across WHO and partner agencies for early warning.

Text embeddings and semantic representation

- **[T]** Mikolov et al. (2013), "Efficient Estimation of Word Representations in Vector Space," arXiv/ICLR Workshop. ? word2vec; foundational distributional word-embedding method, the baseline distributional representation against which sentence-level and transformer embeddings are compared.

- **[T]** Pennington, Socher & Manning (2014), "GloVe: Global Vectors for Word Representation," EMNLP. ? Co-occurrence-matrix-based word embeddings; a further classic baseline.

- **[T]** Devlin et al. (2019), "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL. ? Introduces the masked-language-model transformer architecture underlying nearly all subsequent

sentence-embedding models used in this project.

- **[E]** Reimers & Gurevych (2019), "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," EMNLP. ? Method for deriving fixed-size sentence embeddings efficiently comparable by cosine similarity; the core technique the semantic-matching pipeline will build on.
- **[E]** Karpukhin et al. (2020), "Dense Passage Retrieval for Open-Domain Question Answering," EMNLP. ? Establishes dense embedding retrieval as a replacement for lexical (BM25/TF-IDF) search in open-domain settings; the retrieval paradigm this project transposes to epidemic-intelligence article triage.
- **[E]** Feng et al. (2022), "Language-Agnostic BERT Sentence Embedding" (LaBSE),* ACL. ? Multilingual sentence embedding model aligning ~100 languages in a shared space; a leading candidate for a monitoring workflow that must cover many source languages.
- **[E]** Lee et al. (2020), "BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining,"* Bioinformatics. ? Domain-adapted embeddings for biomedical/veterinary vocabulary; candidate model for domain-fit comparison against general-purpose multilingual embeddings.

Sensitivity, specificity and evaluation without a perfect gold standard

- **[T]** Rogan & Gladen (1978), "Estimating Prevalence from the Results of a Screening Test,"* American Journal of Epidemiology. ? Classic correction of prevalence/accuracy estimates for known test sensitivity and specificity; starting point for propagating calibration-data error through pipeline evaluation.
- **[T]** Hui & Walter (1980), "Estimating the Error Rates of Diagnostic Tests,"* Biometrics. ? Latent-class model for estimating sensitivity and specificity of multiple imperfect tests in the absence of a gold standard; the core statistical framework proposed here for scoring the embedding pipelines against imperfect WAHIS/expert-annotation reference data.
- **[E]** Enøe, Georgiadis & Johnson (2000), "Estimation of Sensitivity and Specificity of Diagnostic Tests and Disease Prevalence When the True Disease State Is Unknown,"* Preventive Veterinary Medicine. ? Applied veterinary-epidemiology treatment of the no-gold-standard estimation problem, directly transposable from diagnostic-test evaluation to event-detection-pipeline evaluation.
- **[E]** Farrington et al. (1996), "A Statistical Algorithm for the Early Detection of Outbreaks of Infectious Disease,"* Journal of the Royal Statistical Society A. ? Reference outbreak-detection algorithm and evaluation approach (false-alarm/detection-delay trade-off), useful as an evaluation-design template alongside the sensitivity/specificity framework.
- **[E]** Yin, Hay & Roth (2019), "Benchmarking Zero-Shot Text Classification: Datasets, Evaluation and Entailment Approach,"* EMNLP. ? Evaluation methodology for embedding-/entailment-based classifiers with no or minimal labelled training data, relevant to how the semantic pipeline will be scored against sparse calibration labels.

Notation du sujet :

Commentaires :